

AM 20 · DISCUSSION SECTION

First-order ODEs, three ways

Modeling · Terminal velocity · Classification

Dante Buhl, Teaching Assistant

WELCOME

Hi, I'm Dante

I'm a PhD student in Applied Mathematics here at UCSC, advised by Pascale Garaud.

I study rotating, stratified turbulence in oceans, atmospheres and stars, using large simulations on supercomputers. Under the hood, that is a lot of differential equations.

More about me: dbuhl8.github.io/website

OFFICE HOURS

Thursdays
4–5 pm

BE 153A

Bring homework questions, lecture questions, or anything else.

HOW SECTION WORKS

You do the math, together



Active learning

A short setup from me, then you work. Most of the hour is you thinking, not me talking.



Group-led solving

Groups of 3–4 work each problem together. Explain your reasoning to each other out loud.



Check-ins every 15 min

We regroup, compare approaches, and get unstuck before moving on.

Problem 1



Check-in



Problem 2



Check-in



Problem 3



Check-in

TODAY

Three problems, picked for Homework 1

#	Problem	Source	Skill you'll practice	HW 1
1	Attendance model	Lecture 1, Problem 14	Solve, then fit an unknown constant to data	Q1, Q2
2	Falling with drag	Lecture 1, Problem 11	Verify a solution; reach a percent of terminal velocity	Q3, Q6, Q7
3	Classifying ODEs	Lecture 2, Problem 10	Order, linearity, and the four classifications	Q4, Q5

The attendance model

Attendance $p(t)$ in AM 20, with t in months, follows

$$\frac{dp}{dt} = \frac{1}{2}p - 100, \quad p(0) = p_0$$

- (a) Find the smallest whole number p_0 for which at least 150 students attend at the end of the quarter, $t = 3$.
- (b) Find p_0 so that attendance reaches zero at exactly $t = 3$.
- (c) Describe what happens for $p_0 = 200$ and for $p_0 > 200$. What does that say about the model?

PROBLEM 1 · KEY IDEAS

Find the equilibrium, then separate

1. Equilibrium first

Set $dp/dt = 0$ to get $p = 200$, then factor it out:

$$\frac{dp}{dt} = \frac{1}{2} (p - 200)$$

2. Separate, integrate

Divide by $p - 200$ and integrate:
 $\ln|p - 200| = t/2 + c$. Exponentiate:

$$p(t) = 200 + C e^{t/2}$$

3. Let data fix constants

$p(0) = p_0$ fixes C . A value at a later time, like $p(3)$, pins down one more unknown.

Same move as finding a in HW Q1 and r in HW Q2.

PROBLEM 1 · SOLUTION

Solve once, then ask three questions

With $C = p_0 - 200$: $p(t) = 200 + (p_0 - 200) e^{t/2}$

(a)

$$\begin{aligned} 150 &= 200 + (p_0 - 200) e^{3/2} \\ p_0 &= 200 - 50 e^{-3/2} \\ &\approx 188.84 \end{aligned}$$

$p(3)$ grows with p_0 , so $p_0 = \mathbf{189}$. Check:
189 gives $p(3) \approx 150.7$; 188 gives
 ≈ 146.2 .

(b)

$$\begin{aligned} 0 &= 200 + (p_0 - 200) e^{3/2} \\ p_0 &= 200 \left(1 - e^{-3/2} \right) \\ &\approx 155.37 \end{aligned}$$

Not a whole number: the model treats p as a continuous quantity.

(c)

$$p_0 = 200 \Rightarrow p(t) \equiv 200$$

For $p_0 > 200$, $C > 0$ and p grows without bound. Solutions leave 200 on both sides, so the equilibrium is **unstable**.

Model limitation: attendance can't exceed enrollment.

Falling with linear drag

An object of mass m is dropped from rest. With $m, g, \gamma > 0$ and v positive downward,

$$m \frac{dv}{dt} = mg - \gamma v, \quad v(0) = 0$$

- (a) Show that $v(t) = \frac{mg}{\gamma} \left(1 - e^{-\gamma t/m}\right)$ solves this initial value problem.
- (b) Find, in terms of m and γ , the time t_{90} when the object reaches 90% of its terminal velocity.
- (c) For the lecture object ($m = 10$ kg, $\gamma = 2$ kg/s), find t_{90} . Does it reach 90% of terminal velocity before landing from 300 m?

PROBLEM 2 · KEY IDEAS

Check by substitution, then take a log

1. Verify, don't solve

Compute the left side and the right side separately, compare, then check $v(0) = 0$.

Same move as HW Q6 and Q7.

2. Terminal velocity

No acceleration means the right side is zero:

$$0 = mg - \gamma v \Rightarrow v_T = \frac{mg}{\gamma}$$

No solving needed.

3. Percent of terminal

$$v(t_{90}) = 0.9 v_T$$

Divide out v_T , isolate the exponential, take $\ln$ of both sides.

HW Q3 asks the same thing at 95%.

PROBLEM 2 · SOLUTION

Both sides match, and t_{90} ignores g

(a)
$$m v'(t) = m \cdot \frac{mg}{\gamma} \cdot \frac{\gamma}{m} e^{-\gamma t/m} = mg e^{-\gamma t/m}$$
$$mg - \gamma v = mg - mg + mg e^{-\gamma t/m} = mg e^{-\gamma t/m}$$
$$v(0) = \frac{mg}{\gamma} (1 - 1) = 0$$

Left side equals right side for every t , and the initial condition holds. So $v(t)$ solves the IVP.

(b)
$$1 - e^{-\gamma t_{90}/m} = 0.9$$
$$e^{-\gamma t_{90}/m} = 0.1$$
$$t_{90} = \frac{m}{\gamma} \ln 10$$

No g : doubling g doubles v_T but leaves t_{90} alone.

(c)
$$t_{90} = 5 \ln 10 \approx 11.51 \text{ s}$$
$$t_g \approx 10.51 \text{ s}$$

It lands about 1 s too early, so **no**. Impact at 43.01 m/s is 0.878 of $v_T = 49 \text{ m/s}$.

Classifying ODEs

Let n be a constant, and consider

$$(a) \quad y'' + t^n y = 0 \qquad (b) \quad y'' + y^n = 0 \qquad (c) \quad y'' + (y')^2 = 0$$

(a) For which values of n is equation (a) linear?

(b) For which values of n is equation (b) linear?

(c) Classify equation (c) under all four classifications: ODE or PDE, scalar or system, order, linear or nonlinear.

PROBLEM 3 · KEY IDEAS

Four questions to ask any equation

Classification	Ask	Answer
ODE or PDE	How many independent variables?	One: ODE. More: PDE.
Scalar or system	How many unknown functions?	One: scalar. More: system.
Order	What is the highest derivative?	Powers on a derivative don't change it.
Linear or not	Does it fit the form below?	If not, it is nonlinear.

$$a_n(t) y^{(n)} + \cdots + a_1(t) y' + a_0(t) y = g(t)$$

y and its derivatives appear to the first power, never inside sin, exp or a power, never multiplied together. Coefficients may depend on t in any way.

PROBLEM 3 · SOLUTION

Where the exponent sits decides it

(a) Every n

$$y'' + \underbrace{t^n}_{\text{coefficient}} y = 0$$

t^n involves only t , so it is just a coefficient. y and y'' appear to the first power. Linear for all n .

(b) $n = 0$ or 1

$$n = 0 : y'' = -1$$

$$n = 1 : y'' + y = 0$$

Now the exponent is on the unknown.
Any other n leaves y^n , which is nonlinear.

(c) Classify

$$y'' + \underbrace{(y')^2}_{\text{nonlinear}} = 0$$

- ODE: one independent variable, t
- Scalar: one unknown, y
- Second order: y'' is highest
- Nonlinear: $(y')^2$
- Bonus: time-invariant

BEFORE YOU GO

Take these moves to Homework 1

HW 1	Move
Q1, Q2	Find the equilibrium, separate, then let the data fix the constant.
Q3	Set $v(t)$ equal to a fraction of v_T and solve for t with a log.
Q6, Q7	Substitute into both sides and compare.
Q4, Q5	Ask the four questions. Powers change linearity, not order.